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Lab assignment 4.1

Q1:

Prompt:#Write a Python function that classifies a given text as Spam or Not Spam

Code:

```
def classify_text(text):
    spam_keywords = ['win', 'free', 'prize', 'click', 'buy now', 'limited time', 'offer', 'winner']
    text_lower = text.lower()

    for keyword in spam_keywords:
        if keyword in text_lower:
            return "Spam"

    return "Not Spam"
# Example usage
texts = [
    "Congratulations! You have won a free lottery ticket."
    "Hello, how are you doing today?",
    "Limited time offer! Click here to buy now.",
    "Don't forget our meeting tomorrow."
]
classifications = {text: classify_text(text) for text in texts}
print(classifications)
# Output: {'Congratulations! You have won a free lottery ticket.': 'Spam', 'Hello, how are you doing today?': 'Not Spam', 'Limited time offer! Click here to buy now.': 'Spam', "Don't forget our meeting tomorrow.": 'Not Spam'}
```

Output:

{'Congratulations! You have won a free lottery ticket.Hello, how are you doing today?': 'Spam',

'Limited time offer! Click here to buy now.': 'Spam',

"Don't forget our meeting tomorrow.": 'Not Spam'}

Q2:prompt:

'''

i am happy today. output:happy

'''

#write a python program that detects the emotion of a sentence and emotions=[happy,sad,angry,excited,nervous,neutral]

```
def detect_emotion(sentence):
    emotions = ['happy', 'sad', 'angry', 'excited', 'nervous', 'neutral']
    sentence = sentence.lower()


    if 'happy' in sentence or 'joy' in sentence or 'pleased' in sentence:
        return 'happy'
    elif 'sad' in sentence or 'unhappy' in sentence or 'sorrow' in sentence:
        return 'sad'
    elif 'angry' in sentence or 'mad' in sentence or 'furious' in sentence:
        return 'angry'
    elif 'excited' in sentence or 'thrilled' in sentence or 'eager' in sentence:
        return 'excited'
    elif 'nervous' in sentence or 'anxious' in sentence or 'worried' in sentence:
        return 'nervous'
    else:
        return 'neutral'
# Example usage
sentence = "i am very sad"
emotion = detect_emotion(sentence)
```

```
print(emotion)
# Output: happy
```

Q3

Prompt:

```
  '''
 90–100 → A
 80–89 → B
 70–79 → C
 60–69 → D
 Below 60 → F
'''
#write a python program that predict a student's grade based on marks.
marks = int(input("Enter the student's marks: "))
if 90 <= marks <= 100:
    grade = 'A'
elif 80 <= marks < 90:
    grade = 'B'
elif 70 <= marks < 80:
    grade = 'C'
elif 60 <= marks < 70:
    grade = 'D'
else:
    grade = 'F'
print(f"The student's grade is: {grade}")
```

Output: Enter the student's marks: 88

The student's grade is: B

Q4:

Prompt:

```
'''
```

March → Mesha

April → Vrishabha

May → Mithuna

June → Karka

July → Simha

August → Kanya

September → Tula

October → Vrischika

November → Dhanu

December → Makara

January → Kumbha

February → Meena

```
'''
#write a python program that predicts a persons zodiac sign based on  month of birth
def zodiac_sign(month):
    zodiac_dict = {
        "March": "Mesha",
        "April": "Vrishabha",
        "May": "Mithuna",
        "June": "Karka",
        "July": "Simha",
        "August": "Kanya",
        "September": "Tula",
        "October": "Vrischika",
        "November": "Dhanu",
        "December": "Makara",
        "January": "Kumbha",
        "February": "Meena"
    }
```

```
    return zodiac_dict.get(month, "Invalid month")
month_input = input("Enter your month of birth: ")
print(f"Your zodiac sign is: {zodiac_sign(month_input)}")
```

output: Enter your month of birth: October

Your zodiac sign is: Vrischika

Q5:

Prompt:

'''

Read the marks from the user.

based on marks determine whether student passed or not.

handle invalid inputs

'''

Code:

```
def check_pass_fail(marks):
    try:
        marks = float(marks)
        if marks < 0 or marks > 100:
            raise ValueError("Marks should be between 0 and 100.")

        if marks >= 40:
            return "Passed"
        else:
            return "Failed"
    except ValueError as e:
        return str(e)
```

```
print("Enter the marks obtained by the student:")
user_input = input()
result = check_pass_fail(user_input)
print(result)
```

Output: 45

Passed

Q6.

Prompt:

'''

read the age of the user

print he can vote or not

handle invalid cases

'''

Code:

```
def can_vote(age):
    try:
        age = int(age)
        if age < 0:
            return "Invalid age. Age cannot be negative."
        elif age >= 18:
            return "You are eligible to vote."
        else:
            return "You are not eligible to vote."
    except ValueError:
        return "Invalid input. Please enter a valid age."
user_age = input("Enter your age: ")
print(can_vote(user_age))
```

Output: Enter your age: 19

You are eligible to vote.

Q7:

Prompt:

'''

Generate a list that named names conists of some names

from list names if the name is palindrome add it to the list palindrome_names

handle invalid cases

'''

```
def is_palindrome(name):
    # Check if the name is a palindrome
    return name == name[::-1]
def get_palindrome_names(names):
    palindrome_names = []
    for name in names:
        if not isinstance(name, str):
            print(f"Invalid name: {name}. Skipping.")
            continue
        if is_palindrome(name):
            palindrome_names.append(name)
    return palindrome_names
names = ["Anna", "Bob", "Cathy", "David", "Eve", 123, None, "Madam"]
palindrome_names = get_palindrome_names(names)
print("Palindrome names:", palindrome_names)
```

Output:

[Anna,bob,David,eve]

Q8:

Prompt:

'''

generate a list of words named words_list

traverse the list and calculate the length of each word.

if length is less than 5 make a list and add as "short" to list

if length greater than 7 add as "long" to list

'''

Code:

```
words_list = ["apple", "banana", "kiwi", "strawberry", "fig", "watermelon", "pear", "grape"]
length_category = []
for word in words_list:
    if len(word) < 5:
        length_category.append("short")
    elif len(word) > 7:
        length_category.append("long")
print(length_category)
```

Output: ['short', 'long', 'short', 'long', 'short']